Jenkins challenges 2

Create a Declarative Jenkins parameterized pipeline for simplecustomerapp.

Requirements:

- Create a Pipeline job in Jenkins.
- Use a Git repository for simplecustomerapp.
- Add parameters:
  - BRANCH_NAME (string, default: main)
  - ENV (choice: dev, qa)
- Define environment variables:
  - APP_NAME = simplecustomerapp
- Implement stages:
  - Checkout (based on BRANCH_NAME)
  - Build (mvn clean compile)
  - Test (mvn test)
  - Package (mvn package)
- Add post actions:
  - Success → print success message with environment
  - Failure → print failure message
- Trigger using Build with Parameters.

Outcome

- Parameterized Jenkins pipeline created
- Code is built, tested, and packaged automatically
- JAR file generated in target/
- Clear visibility of build stages and logs
- Single pipeline works for multiple environments

Advantages

- One pipeline can be reused for different branches and environments
- Reduces manual build and deployment effort
- Early test failures stop bad code from moving forward
- Easy to maintain and extend (SonarQube, Docker, Deploy stages later)

Prerequisites (Before Starting)

Before creating the Jenkins pipeline, make sure the following are **already available**:

1) Jenkins Server is Running

- Jenkins should be **installed and running**
- Jenkins UI should be accessible in browser
  Example:

```
http://<jenkins-ip>:8080
```

-

2) Java Installed on Jenkins Server

- Jenkins needs Java to run Maven builds

Check:

```
java -version
```

Expected:

```
Java 8 or Java 11
```

3) Maven Installed on Jenkins Server

- Required to build and package the JAR file

Check:

```
mvn -version
```

4) Git Installed on Jenkins Server

- Needed to clone the Git repository

Check:

```
git --version
```

5) Internet Access from Jenkins Server

- Jenkins must be able to access GitHub:

```
https://github.com/Waseema761/SimpleCustomerApp.git
```

6) Required Jenkins Plugins Installed

Go to:
Manage Jenkins → Plugins

Make sure these are installed:

- ✔ Git Plugin
- ✔ Pipeline Plugin
- ✔ Pipeline: Declarative

7) Git Repository Ready

- Repository contains:
  - `pom.xml`
  - `src/main/java`
  - Maven project structure

Repo used:

```
SimpleCustomerApp
```

Steps to Create Parameterized Declarative Pipeline

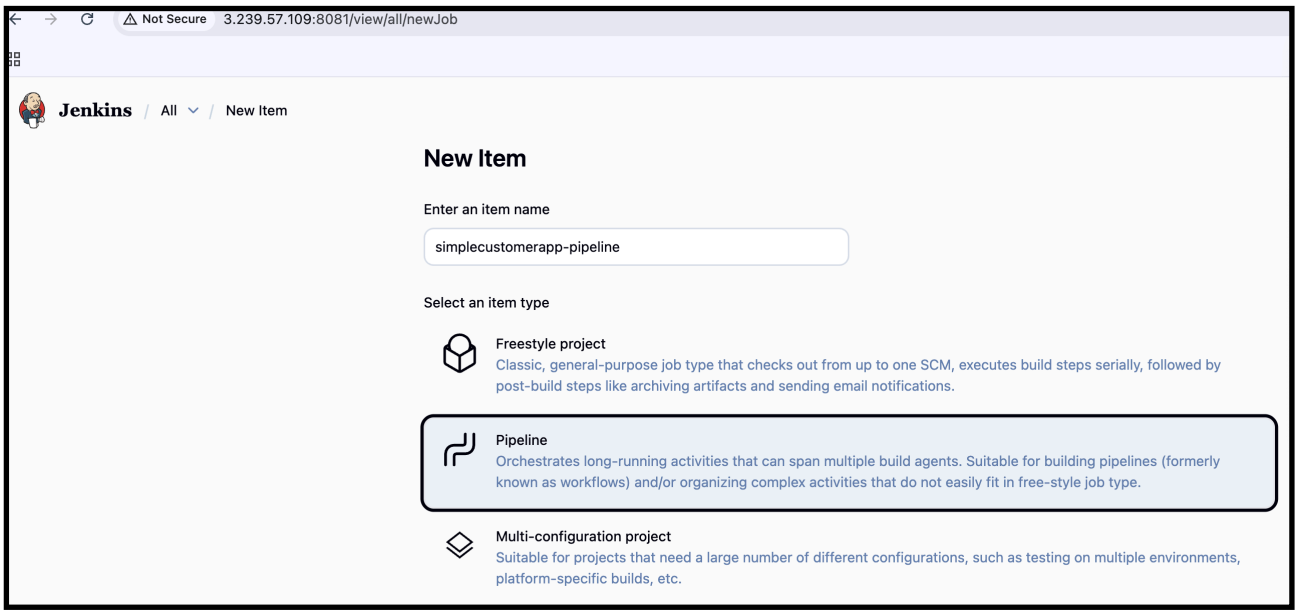
Step 1. Create a New Pipeline Job

1. Open **Jenkins Dashboard**
2. Click **New Item**
3. Enter Job Name:

```
simplecustomerapp-pipeline
```

4.

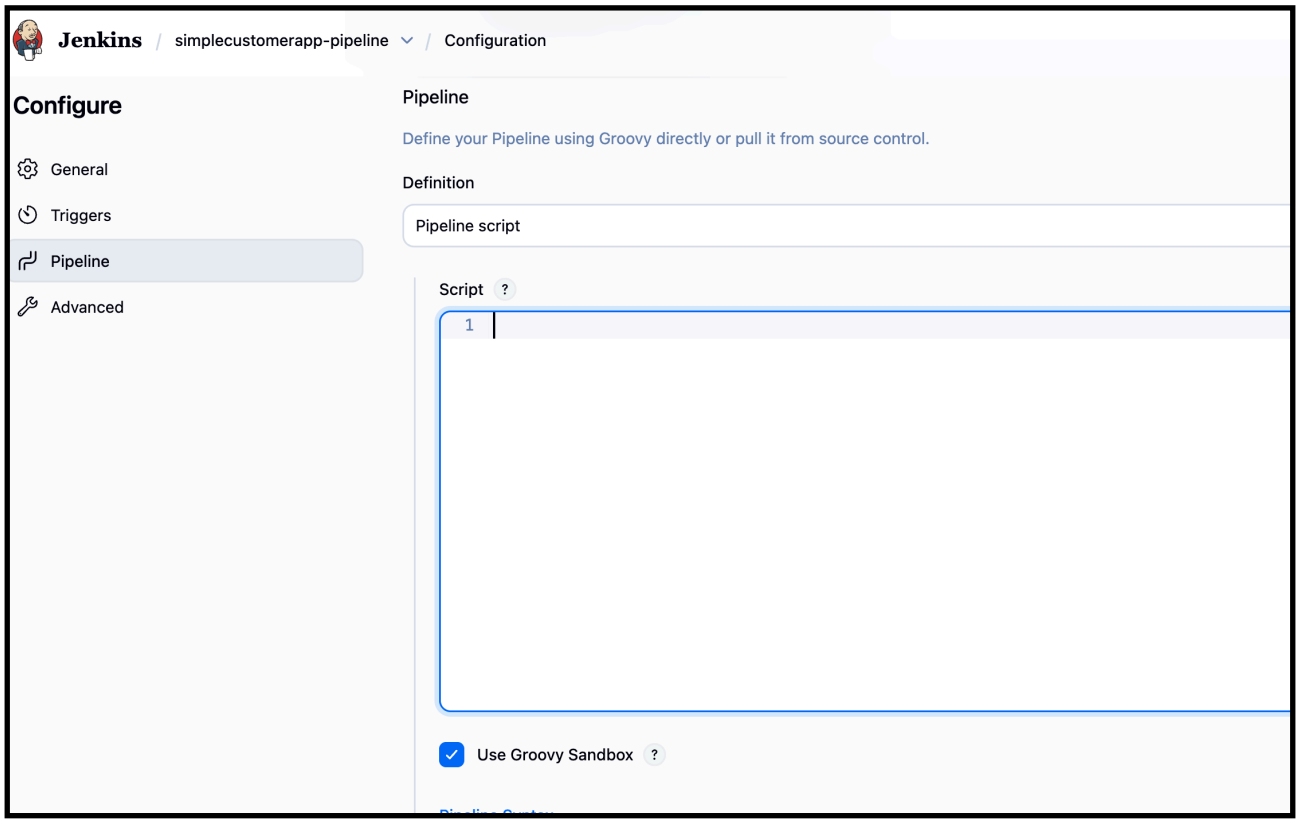
5. Select **Pipeline**
6. Click **OK**



Step 2. Configure Pipeline

1. Scroll to **Pipeline** section
2. Choose:

```
Definition → Pipeline script
```



Paste the Declarative Jenkins pipeline code: below

```
pipeline {
    agent any

    parameters {
        string(name: 'BRANCH_NAME', defaultValue: 'master')
        choice(name: 'ENV', choices: ['dev', 'qa'])
    }

    environment {
        APP_NAME = 'simplecustomerapp'
    }

    stages {

        stage('Checkout') {
            steps {
                git branch: "${params.BRANCH_NAME}",
                    url: 'https://github.com/Waseema761/
SimpleCustomerApp.git'
            }
        }

        stage('Verify Workspace') {
            steps {
                sh '''
                    pwd
                    ls -l
                    echo "---- inside SimpleCustomerApp ----"
                    cd SimpleCustomerApp
                    pwd
                    ls -l
                    ls -l pom.xml
                '''
            }
        }

        stage('Build') {
            steps {
                dir('SimpleCustomerApp') {
                    sh 'mvn clean compile'
                }
            }
```

```
        }

        stage('Test') {
            steps {
                dir('SimpleCustomerApp') {
                    sh 'mvn test'
                }
            }
        }

        stage('Package') {
            steps {
                dir('SimpleCustomerApp') {
                    sh 'mvn package'
                }
            }
        }
    }

    post {
        success {
            echo "BUILD SUCCESS for ${APP_NAME} in ${params.ENV}"
            sh 'ls -l SimpleCustomerApp/target/'
        }
        failure {
            echo " BUILD FAILED for ${APP_NAME}"
        }
    }
}
```

Step:3

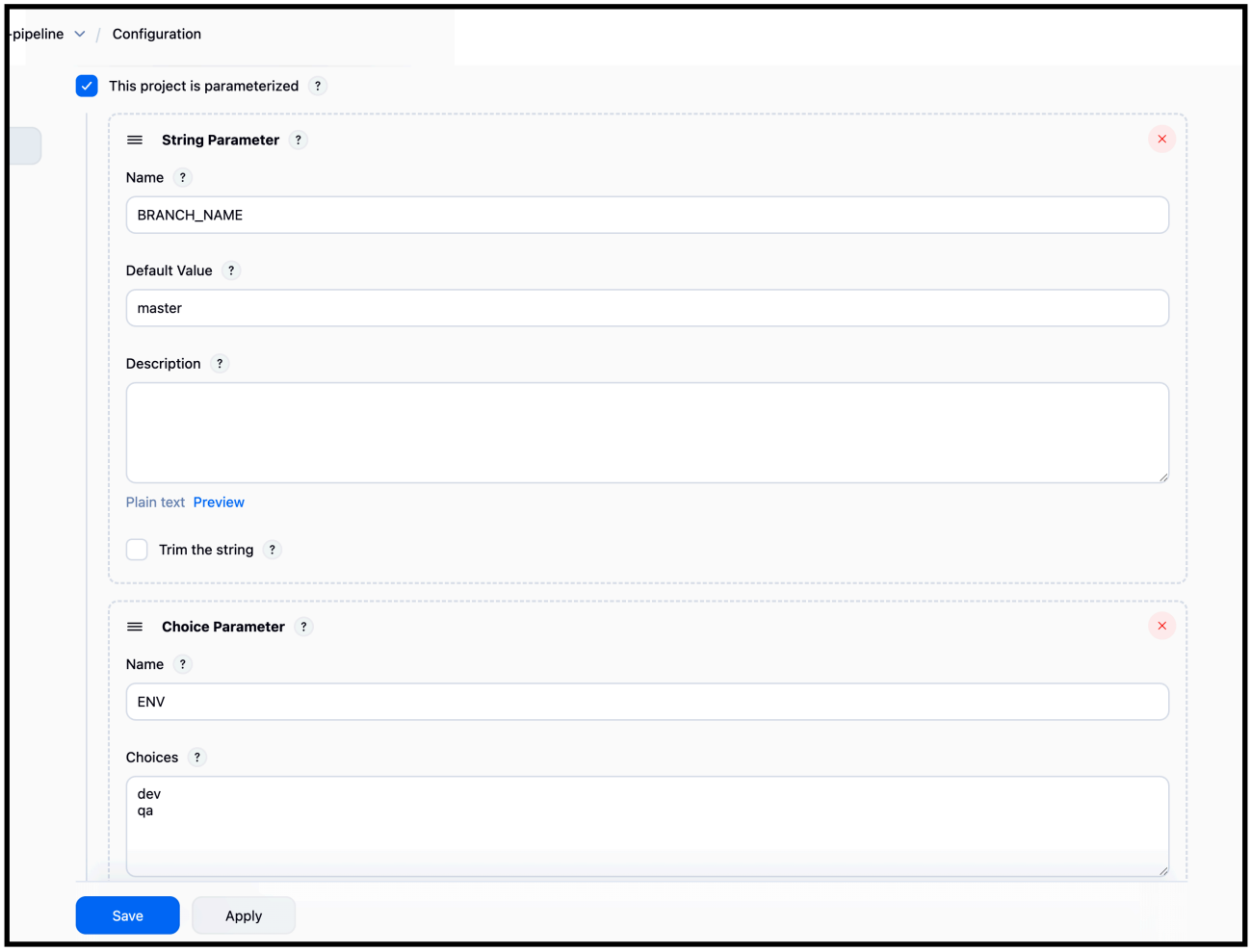
Add Pipeline Parameters

The pipeline is parameterized to support multiple branches and environments.

Parameters used:

- **BRANCH_NAME** (String)
    - Default value: `master`
    - Used to select the Git branch to build

- **ENV** (Choice)
  - Values: `dev`, `qa`
  - Used to represent the target environment



Step 4: Define Environment Variable

An environment variable is defined inside the pipeline:

`APP_NAME = simplecustomerapp`
This variable is used in log messages and post actions.

```
8
9       environment {
10          APP_NAME = 'simplecustomerapp'
11      }
12
```

Step 5: Implement Pipeline Stages

5.1 Checkout Stage

- Clones the Git repository
- Checks out the branch provided in `BRANCH_NAME` parameter

Purpose:

- Fetch application source code from GitHub

```
13    stages {
14
15        stage('Checkout') {
16            steps {
17                git branch: "${params.BRANCH_NAME}",
18                    url: 'https://github.com/Waseema761/SimpleCustomerApp.git'
19            }
20        }
21
```

5.2 Build Stage

- Runs Maven build command:

```
mvn clean compile
```

-

Purpose:

- Compiles the Java source code
- Ensures there are no compilation errors

```
36        stage('Build') {
37            steps {
38                dir('SimpleCustomerApp') {
39                    sh 'mvn clean compile'
40                }
41            }
42        }
43
```

5.3 Test Stage

- Runs Maven test command:

```
mvn test
```

-

Purpose:

- Executes unit tests
- Ensures code quality before packaging

```
43
44          stage('Test') {
45              steps {
46                  dir('SimpleCustomerApp') {
47                      sh 'mvn test'
48                  }
49              }
50          }
51
```

5.4 Package Stage

- Runs Maven package command:

```
51
52          stage('Package') {
53              steps {
54                  dir('SimpleCustomerApp') {
55                      sh 'mvn package'
56                  }
57              }
58          }
59      }
```

```
mvn package
```

-

Purpose:

- Packages the application into a **JAR file**
- JAR file is generated inside the `target/` directory
-

Step 6: Post Actions

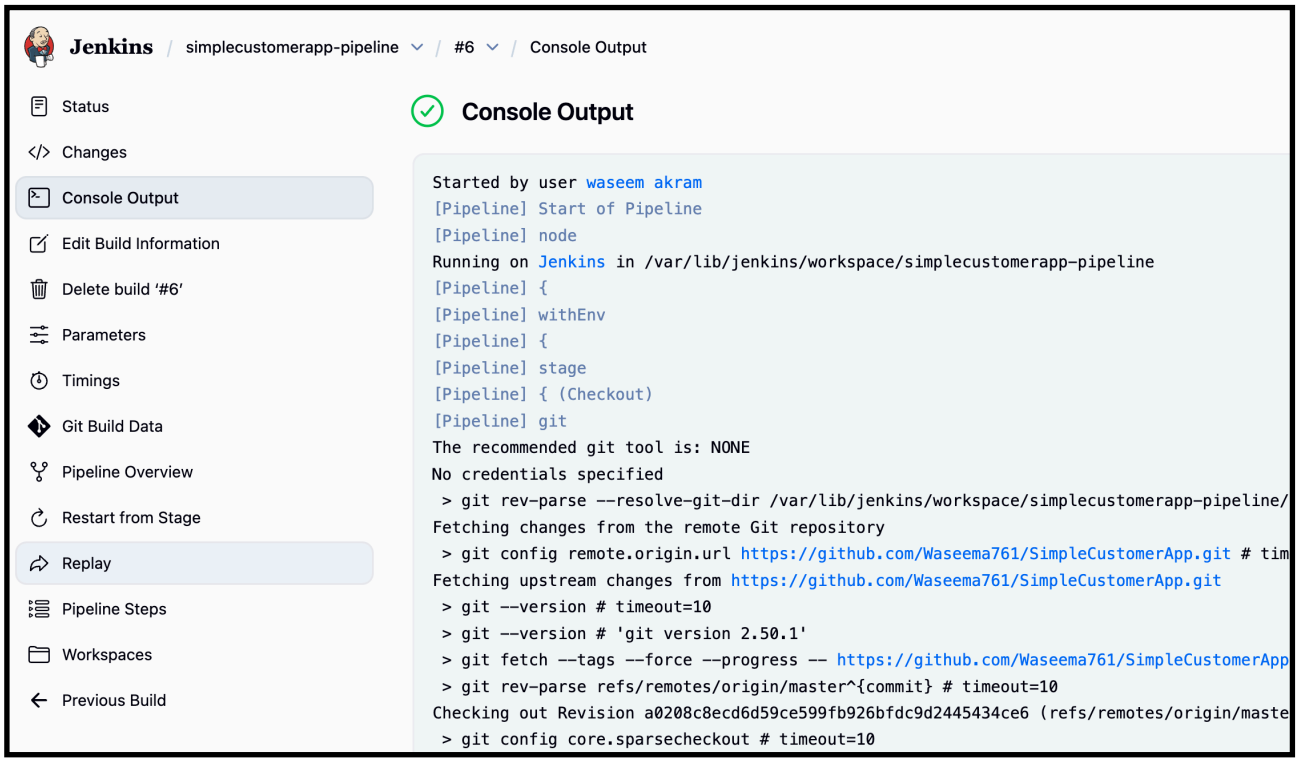
—-> now build the code click build now and check the success /failure

Success Action

- Prints success message with environment name
- Displays the generated JAR file location

Example:

```
BUILD SUCCESS for simplecustomerapp in dev environment
```



Output Verification:

After successful pipeline execution, the JAR file is generated at:

`/var/lib/jenkins/workspace/simplecustomerapp-pipeline/SimpleCustomerApp/target/`

```
[root@ip-172-31-65-24 ~]# cd /var/lib/jenkins/workspace/
[root@ip-172-31-65-24 workspace]# ls
'MULTI-STAGE PIPELINE'       hiring-app-declarative       job2                    pipeline      'scripted pipelin
'MULTI-STAGE PIPELINE@tmp'   hiring-app-declarative@tmp   job3                    pipeline@tmp  'scripted pipelin
 first-job                   hiring-app-parallel          parametirized-job       private        scripted-pipelin
 hiring-app                  hiring-app-parallel@tmp      parametirized-job@tmp   private@tmp    scripted-pipelin
[root@ip-172-31-65-24 workspace]# cd simplecustomerapp-pipeline
[root@ip-172-31-65-24 simplecustomerapp-pipeline]# ls
README.md  SimpleCustomerApp  SimpleCustomerApp@tmp
[root@ip-172-31-65-24 simplecustomerapp-pipeline]# cd simplecustomerAPP
-bash: cd: simplecustomerAPP: No such file or directory
[root@ip-172-31-65-24 simplecustomerapp-pipeline]# cd SimpleCustomerApp
[root@ip-172-31-65-24 SimpleCustomerApp]# ls
Service-API.dockerfile  WebContent  XMLSave.txt  build  build.xml  pom.xml  sonar-project.properties  src  target
[root@ip-172-31-65-24 SimpleCustomerApp]# cd target
[root@ip-172-31-65-24 target]# ld
ld: no input files
[root@ip-172-31-65-24 target]# ls
SimpleCustomerApp-1.0-SNAPSHOT.jar  maven-archiver
[root@ip-172-31-65-24 target]# cd
```

In target we are able to see

simpleCustomerApp-1.0-SNAPSHOT.jar

Console output :

```
-rw-r--r--. 1 jenkins jenkins 461 Jan  7 07:27 pom.xml
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] dir
Running in /var/lib/jenkins/workspace/simplecustomerapp-pipeline/SimpleCustomerApp
[Pipeline] {
[Pipeline] sh
+ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.javatpoint:SimpleCustomerApp >------------------
[INFO] Building SimpleCustomerApp 1.0-SNAPSHOT
[INFO]   from pom.xml
```

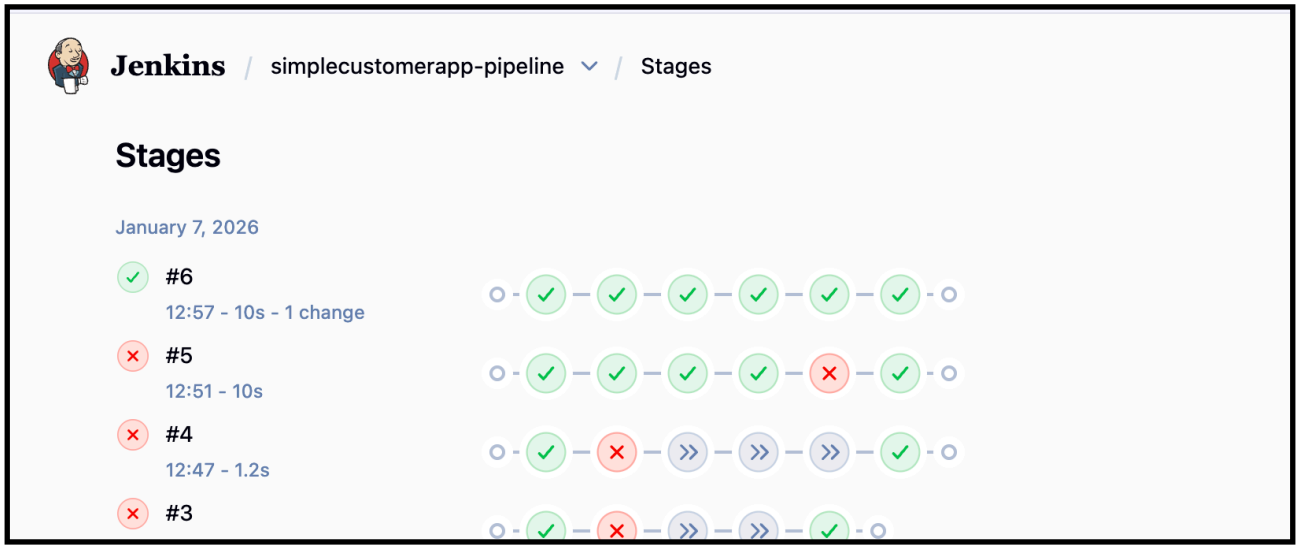
```
Running in /var/lib/jenkins/workspace/simplecustomerapp-pipeline/SimpleCustomerApp
[Pipeline] {
[Pipeline] sh
+ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.javatpoint:SimpleCustomerApp >-------------------
[INFO] Building SimpleCustomerApp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ SimpleCustomerApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/simplecustomerapp-pipeline/SimpleCustomerApp/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ SimpleCustomerApp ---
[INFO] No sources to compile
[INFO]
```

From pom.xml building jar

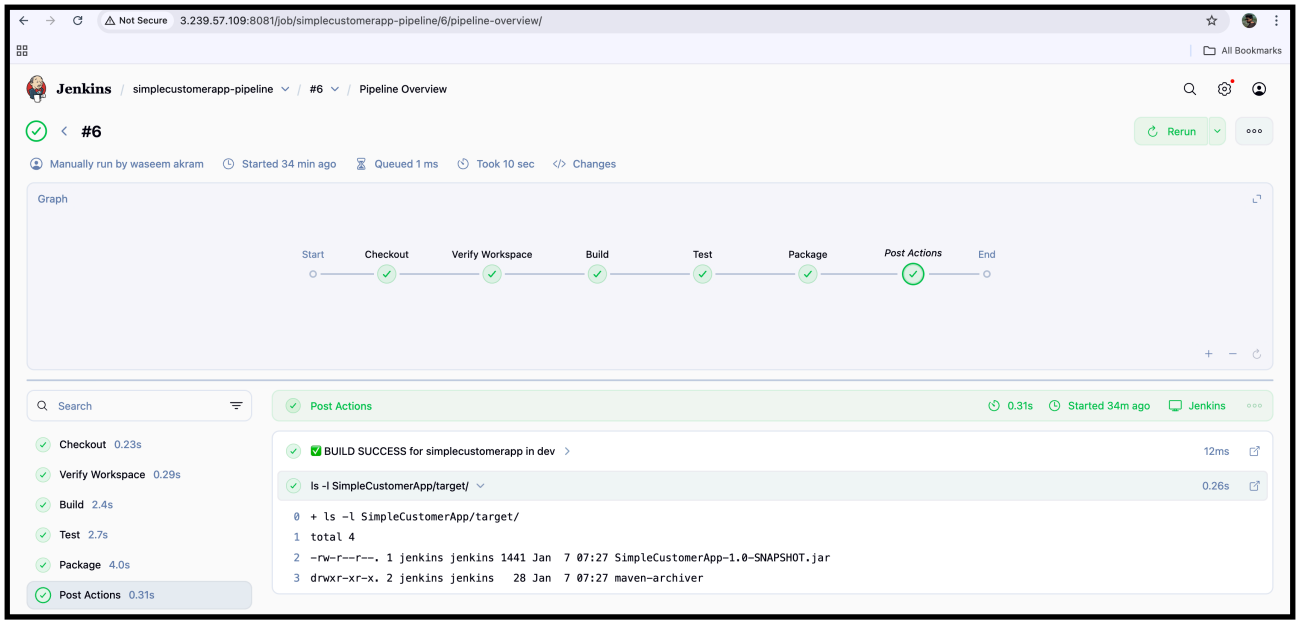
```
[Pipeline] echo
✅ BUILD SUCCESS for simplecustomerapp in dev
[Pipeline] sh
+ ls -l SimpleCustomerApp/target/
total 4
-rw-r--r--. 1 jenkins jenkins 1441 Jan  7 07:27 SimpleCustomerApp-1.0-SNAPSHOT.jar
drwxr-xr-x. 2 jenkins jenkins   28 Jan  7 07:27 maven-archiver
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

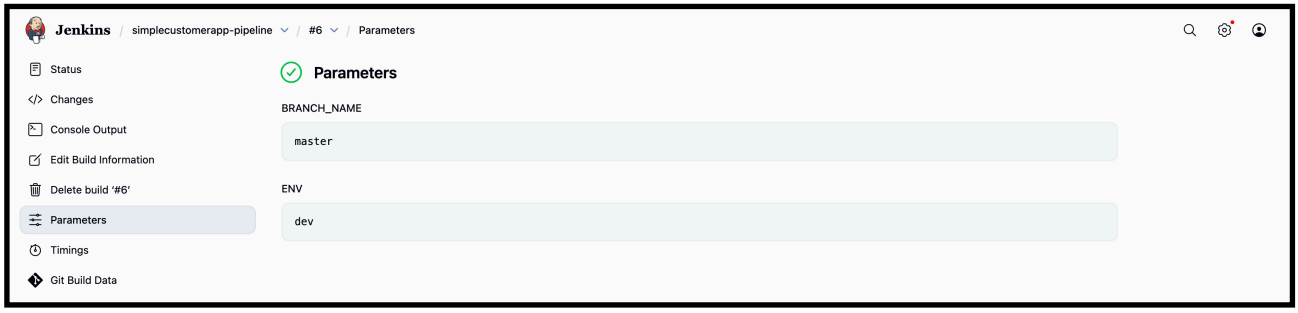
Build succes

Stages :



—> pipeline overview :





Outcome:

- Parameterized Declarative Jenkins pipeline created
- Code is built, tested, and packaged automatically
- JAR file successfully generated
- Clear visibility of pipeline stages and logs
- Same pipeline works for multiple environments

Conclusion:

This task demonstrates the creation of a real-world Jenkins Declarative Parameterized Pipeline.
The pipeline automates code checkout, build, testing, and packaging, and successfully generates a JAR file using Maven.